LASSO, ridge and aggregation methods

Guillaume Lecué*

1 Notation and models

We present here the input/output problem and two probabilistic modelings of this problem. An input/output problem is a prediction problem based on a set of n observations of the type $\{(Y_1, X_1), \ldots, (X_n, Y_n)\}$. The variables $Y_1, \ldots, Y_n$ are generally real numbers (for example the insulin level in the blood, and so on) or binary (SPAM or MAIL, and so on); these variables are called the *outputs* in general, and *labels* in the case of a binary output. The variables $X_1, \ldots, X_n$ are called the *inputs* and may be elements of any abstract space $\mathcal{X}$. The aim is to explain the outputs in terms of the inputs and, in particular, to be able, for a new input X, to predict the associated value Y.

So far no random aspect has been introduced into the problem. The probabilistic aspect appears only at the *modeling* stage (and only if one chooses a probabilistic modeling of the model!). Modeling is an important step in the study of concrete problems. It is this step that translates a problem based on real data into a mathematical problem. Here we shall model this input/output prediction problem by three different models. In all three cases the outputs $Y_1, \ldots, Y_n$ are random. This expresses the idea that what one observes in Y is a noisy version of a function of the input X. Since knowledge of the origins of the noise is "uncertain" or "unknown", one tends to model this uncertainty by a random phenomenon. For two of the models, we shall also assume that the inputs $X_1, \ldots, X_n$ are random variables. This likewise expresses a certain form of uncertainty that one may have about the process generating the inputs. In the third model, one knows the way in which the inputs are created, and one therefore models them in a deterministic way (that is, nonrandom!).

Once the modeling is complete, and the mathematical problem has therefore been posed (corresponding to the original concrete problem, but in mathematical terms), one can carry out simulations and prove theorems, and even develop new theories. The two subsections that follow therefore present the different models used afterwards for this input/output problem and set out the associated mathematical models. The rest of the lecture notes presents methods for solving these mathematical problems.

1.1 Two linear input/output models with random design

We consider a linear input/output problem: one observes n pairs (X_i, Y_i) where $Y_i \in \mathbb{R}$, $X_i \in \mathbb{R}^p$, and one wishes to model a relation of the type

$$Y_i \simeq X_i' \beta \tag{1}$$

where $\beta \in \mathbb{R}^p$. The observations $Y_1, \ldots, Y_n$ are called the *outputs* and are real-valued; the $X_1, \ldots, X_n$ are called the *inputs*, or the *design*, and take values in a space with p variables.

*ESSEC, Business school. Email: lecue@essec.edu.

In general, the inputs may take their values in arbitrary abstract spaces (sets of images, emails, and so on). Nevertheless, in order to emphasize the difference between the case of random-design models (which will serve as the framework for the study of aggregation methods) and deterministic designs (which will serve as the framework for the study of methods of empirical-risk penalization), we restrict ourselves to the study of inputs taking values in $\mathbb{R}^p$.

One can model this problem in a probabilistic way by saying that $(X_1, Y_1), \dots, (X_n, Y_n)$ are n i.i.d. realizations of one and the same random variable (X, Y). In this case, the approximation (1) means that one believes (a priori) that the output variable Y may be a linear function of the *covariates*, that is, of the coordinates of the input variable X. One will therefore seek to estimate Y by linear functions of X, that is, of the form $\langle X, \hat{\beta} \rangle$ for some $\hat{\beta} \in \mathbb{R}^p$ constructed from the data $\mathcal{D} := \big\{(X_1, Y_1), \dots, (X_n, Y_n)\big\}$. In what follows, we shall call $\hat{\beta}$ an estimator, with the understanding that Y will be estimated by $\langle X, \hat{\beta} \rangle$.

We shall rely on the following two statistical models of our data. These two models are probabilistic but of different inspiration. The first is purely statistical (that is, a statistical model is assumed to produce the data). The second is of "computer science" inspiration, in which one does not wish to make a strict assumption on the data, and therefore no statistical model is assumed at the beginning of the analysis.

1. **the "statistical" setting**: one assumes that the data $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. with the same (unknown) distribution as (X, Y) and, moreover, that (X, Y) satisfies a statistical model. In our case (that is, when one believes that a certain linear relation (1) between input and output holds), one assumes that (X, Y) is such that, for some $\beta^* \in \mathbb{R}^p$,

$$Y = \langle X, \beta^* \rangle + \varepsilon \tag{2}$$

   where $\mathbb{E}(\varepsilon | X) = 0$. Concretely, (2) means that one assumes that the regression function of Y given X is a linear function of X (i.e. $\mathbb{E}\big(Y|X\big) = \langle X, \beta^* \rangle$). Then, in order to recover the model (2), it suffices to set $\varepsilon := Y - \mathbb{E}(Y|X)$.

   There is a fairly rich terminology surrounding the model (2). In particular, (2) is called the *linear model* because the regression function is linear in X. It is a *parametric* model because the regression function depends only on a finite number of parameters. Moreover, (2) is called a *Gaussian linear model* when ε is a Gaussian random variable. The variable ε is called the *noise*. When the variance of the noise is independent of X (i.e. $\mathbb{E}\big(\epsilon^2 | X\big) = \sigma^2$ for some constant σ^2), one speaks of *homoscedastic* noise, or of a homoscedastic model (in the opposite case one speaks of *heteroscedasticity*). It is also common to assume that the noise is bounded (i.e. $|\varepsilon| \leq b$ a.s.); one then speaks of *bounded linear regression*.

   The data set $\mathcal{D}$ is used for the construction of statistics (measurable functions of the data). In this model, we focus on linear statistics, that is, estimators of the form $\langle \cdot, \hat{\beta} \rangle$ where $\hat{\beta} = \hat{\beta}(\mathcal{D})$ is a vector constructed from the data $\mathcal{D}$. There are different ways of quantifying the quality of the estimation of Y by an estimator of the form $\langle \cdot, \beta \rangle$ for a fixed $\beta \in \mathbb{R}^p$. For example, one may consider the quadratic prediction risk:

$$R(\beta) := \mathbb{E}_{(X,Y)\sim P}\left[\big(Y - X'\beta\big)^2\right].$$

   There also exist estimation risks: $\|\hat{\beta} - \beta^*\|_p$ for some $p \geq 1$, or else $\mathbb{E}\big(\langle X, \hat{\beta} - \beta^* \rangle^2\big)$.

2. **the "learning theory" setting**: the pairs (X_i, Y_i) are i.i.d. with unknown distribution, distributed as (X, Y), and one assumes nothing more. In particular, one does not assume

that (X, Y) satisfies a certain relation as in the previous "statistical" setting. In this case, the goal is to find a parameter $\beta \in \mathbb{R}^p$ minimizing the risk

$$R(\beta) = \mathbb{E}_{(X,Y)\sim P}\left[\left(Y - X'\beta\right)^2\right].$$

We shall denote by $\beta^* \in \mathbb{R}^p$ a vector such that $R(\beta^*) = \min_{\beta\in\mathbb{R}^p} R(\beta)$, that is,

$$\beta^* \in \operatorname{Arg}\min_{\beta\in\mathbb{R}^p} R(\beta).$$

By definition, $\left\langle X, \beta^* \right\rangle$ is the best (in the L_2 sense) linear estimator of Y. The vector β^* is sometimes called an *oracle* because it is a vector to which one does not have access from the data alone, and which one would like to know.

Given an estimator $\hat{\beta} = \hat{\beta}(X_1, Y_1, ..., X_n, Y_n) = \hat{\beta}(\mathcal{D})$, one will judge the relevance of the associated linear estimator $\left\langle \cdot, \beta \right\rangle$ in view of the risk difference $R(\hat{\beta}) - R(\beta^*)$ (which is always nonnegative). In other words, the smaller $R(\hat{\beta}) - R(\beta^*)$ is, the better the estimation of Y by the linear estimator $\left\langle X, \hat{\beta} \right\rangle$. In this case, the risk of an estimator is the random variable (a function of the data set $\mathcal{D}$):

$$R(\hat{\beta}) := \mathbb{E}_{(X,Y)\sim P}\big[(Y - X'\hat{\beta})^2 | \mathcal{D}\big].$$

Since the distribution P is unknown, the function $R(\cdot)$ is unknown as well; one does, however, have at one's disposal the empirical risk function

$$r(\beta) := \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - X_i'\beta\right)^2.$$

For a fixed $\beta \in \mathbb{R}^p$, one hopes that the empirical quantity (that is, constructed from what one observes) $r(\beta)$ is a good approximation of the theoretical risk $R(\beta)$ (which is not observable and which one would like to know). Indeed, "on average" one has $\mathbb{E}r(\beta) = R(\beta)$, and by the strong law of large numbers one has $r(\beta) \to R(\beta)$ as n tends to infinity. Thus, asymptotically (that is, as n tends to infinity), $r(\beta)$ will be close to $R(\beta)$. But, from a statistical point of view, "asymptotically" means that the statistician has an infinite amount of data, which is never the case. One would therefore like $r(\beta)$ to be close to $R(\beta)$ for every value of n (the number of observations). This kind of problem is related to concentration inequalities, which ensure that "for every ϵ, there exists $0 < \delta_n(\epsilon) < 1$ such that, with probability greater than $1 - \delta_n(\epsilon)$, $|r(\beta) - R(\beta)| \leq \epsilon$". This kind of inequality is fundamental in statistical learning theory (or "machine learning"). Finally, when the output variable Y takes only two values, for example $Y \in \{0, 1\}$, this model is the *classification* model (or "pattern recognition").

These two settings are studied with rather different theoretical tools. The estimators proposed in the two cases are, however, in general the same. We shall therefore present the estimators in a fairly general way; we shall then give results in setting (1) (easier to study), but one should know that they can also be studied in setting (2) with the usual tools of that setting: concentration inequalities, complexity calculations, and a few geometric inequalities.

1.2 A linear input/output model with deterministic design

There is another probabilistic way of modeling the input/output problem defined in the previous section, which consists in assuming that the inputs $X_1, \ldots, X_n$ are deterministic (i.e. nonrandom).

This model is statistical in the sense that one assumes that the data satisfy an equation of the form, for some $\beta^* \in \mathbb{R}^p$,

$$Y_i = \langle X_i, \beta^* \rangle + \varepsilon_i, \quad i = 1, \dots, n \tag{3}$$

where $\varepsilon_1, \dots, \varepsilon_n$ are n random variables such that $\mathbb{E}\varepsilon_i = 0$ for every $i = 1, \dots, n$. In this case one speaks of a *linear regression model with deterministic design*, or of a *linear denoising model*. The latter terminology comes from the fact that one may imagine that $X_1, \dots, X_n$ have been fixed in advance (for example according to an *experimental design*) and that one observes the function $\langle \cdot, \beta^* \rangle$ in a noisy way at the points $X_1, \dots, X_n$. One therefore wishes to "denoise" the observations $Y_1, \dots, Y_n$, that is, to estimate the values $\langle X_i, \beta^* \rangle$ for every $i = 1, \dots, n$. For an estimator $\hat{\beta}$, one will therefore be interested in a control of the quantity

$$\frac{1}{n} \sum_{i=1}^{n} \left(\langle X_i, \beta^* \rangle - \langle X_i, \hat{\beta} \rangle \right)^2$$

with high probability, that is, a result of the type: "with high probability, $n^{-1} \sum_{i=1}^n \left(\langle X_i, \beta^* \rangle - \langle X_i, \hat{\beta} \rangle \right)^2$ is smaller than a given ϵ", or in expectation: "$\mathbb{E} n^{-1} \sum_{i=1}^n \left(\langle X_i, \beta^* \rangle - \langle X_i, \hat{\beta} \rangle \right)^2 \leq \epsilon$". In this framework, one may also be interested in the estimation of β^*. One will then seek to construct estimators $\hat{\beta}$ for which $\left\| \beta^* - \hat{\beta} \right\|_p$ is small for some $p \geq 1$ (in probability and/or in expectation).

In this framework, the empirical risk is still given, for every vector $\beta \in \mathbb{R}^p$, by

$$r(\beta) = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \langle X_i, \beta \rangle)^2.$$

1.3 Notation

By convention, every vector v in $\mathbb{R}^d$ is a column vector; we denote by v' its transpose. We therefore have $v = (v_1, ..., v_d)'$. We shall write, for every $q > 0$,

$$\|v\|_q = \left(\sum_{i=1}^{d} |v_i|^q \right)^{\frac{1}{q}} \text{ and } \|v\|_\infty = \max_{1 \leq i \leq q} |v_i|,$$

and by extension

$$\|v\|_0 = \sum_{i=1}^{d} \mathbb{1}_{\{v_i \neq 0\}}.$$

The normalized norms of $\mathbb{R}^d$ are denoted, for every $v \in \mathbb{R}^d$, by

$$\|v\|_{L_q^n} = \left(\frac{1}{d} \sum_{i=1}^{d} |v_i|^q \right)^{1/q}.$$

The renormalized inner product is defined, for every $u, v \in \mathbb{R}^p$, by

$$\langle v, u \rangle_n = \frac{1}{n} \sum_{i=1}^{d} u_i v_i.$$

We shall denote by I_d the identity matrix of size d and, for every matrix A, by A' its transpose and by $\mathrm{Tr}(A)$ its trace.

2 Methods of empirical-risk penalization

The study of these methods will be carried out in the framework of the **linear model with deterministic design**. For greater simplicity in what follows, we shall assume that the design has been renormalized, that is, that the matrix $x'x/n$ has only 1's on the diagonal, and centered, that is, that $(1, ..., 1)x = (0, ..., 0)$. We shall study four estimators in what follows. The first is the most natural in the framework of the denoising problem, since it consists in seeking, among all vectors $\beta \in \mathbb{R}^p$, the one with the smallest empirical risk — called the least-squares estimator. We shall see that this estimator behaves very well as long as p (the dimension of the input variables) remains smaller than the number n of observations. In the opposite case — when p is larger than n — one speaks of a *high-dimensional problem*. In this framework, the least-squares estimator is no longer efficient. One then proposes to study estimators based on the penalization of the empirical risk. These are estimators of the form

$$\hat{\beta} \in \arg\min_{\beta \in \mathbb{R}^p} \big(r(\beta) + \mathrm{pen}(\beta)\big)$$

where pen is a function on $\mathbb{R}^p$ that penalizes the vectors one wishes to "avoid" (this term will be made clearer in what follows). In particular, we shall study the penalties $\mathrm{pen}(\beta) = \|\beta\|_0, \|\beta\|_1$ and $\|\beta\|_2$.

2.1 Least-squares estimator

The least-squares estimator (Ordinary Least Square – OLS – estimator), also called the empirical risk minimizer, is defined by

$$\hat{\beta}_{OLS} = \arg\min\{r(\beta), \beta \in \mathbb{R}^p\} = \arg\min\{\|y - x\beta\|_2^2, \beta \in \mathbb{R}^p\}.$$

This minimum is unique if and only if the matrix x has rank p (or, equivalently, $x'x$ is invertible), and in that case one has the following expression

$$\hat{\beta}_{OLS} = (x'x)^{-1}x'y.$$

Indeed, one can easily see that the function $F : \beta \to \|y - x\beta\|_2^2$ is convex and therefore attains its minimum at $\hat{\beta}_{OLS}$ such that $\nabla F(\hat{\beta}_{OLS}) = 0$, that is, $-2x'(y - x\hat{\beta}_{OLS}) = 0$.

The condition "x has rank p" necessarily entails $p \leq n$. That is, there are more observations than parameters. One then has the following result on the estimation performance of the least-squares estimator.

Theorem 2.1 *We place ourselves in the setting of linear regression with deterministic design introduced in Subsection 1.2: we are given $X_1, \ldots, X_n$, deterministic vectors in $\mathbb{R}^p$, and we observe, for some $\beta^* \in \mathbb{R}^p$,*

$$Y_i = \langle X_i, \beta^* \rangle + \varepsilon_i$$

where the ε_i are independent, centered ($\mathbb{E}(\varepsilon_i) = 0$) and of variance σ^2 (i.e. $\mathbb{E}(\varepsilon_i^2) = \sigma^2$). Then one has

$$\mathbb{E}\left(\left\|x(\hat{\beta}_{OLS} - \beta^*)\right\|_2^2\right) = p\sigma^2.$$

We denote the risk in the linear denoising model by

$$R(\beta) = \frac{1}{n}\sum_{i=1}^{n}(\langle X_i, \beta \rangle - \langle X_i, \beta^* \rangle)^2.$$

The previous result therefore translates in terms of risk as

$$\mathbb{E}\left[R(\hat{\beta}_{OLS}) - R(\beta^*)\right] = \frac{p\sigma^2}{n}.$$

One sees here that the estimation of β^*, which is equivalent to the estimation of its p coordinates $\beta_i^*, i = 1, \ldots, p$, is achieved at the rate $p\sigma^2/n$. This is a rate of the form

$$\frac{(\text{number of parameters to estimate}) \times (\text{variance of the noise})}{\text{number of observations}}.$$

One can understand this rate of convergence in the following way: for each parameter to be estimated, one pays the price σ^2/n. Thus, the more parameters there are to estimate, that is, the larger the dimension of the input space, the more costly the estimation. In particular, one must have $p < n$ in order to have an "efficient" estimation of β^*. We shall see later, however, that there exist estimation procedures in the case $p >> n$. The framework $p >> n$ (more parameters than observations) is called a *high-dimensional problem*. For the moment, we give a proof of Theorem 2.1.

Proof of Theorem 2.1: One has

$$\begin{aligned} x(\hat{\beta}_{OLS} - \beta^*) &= x\left[(x'x)^{-1}x'y - \beta^*\right] \\ &= x\left[(x'x)^{-1}x'x\beta^* - (x'x)^{-1}x'\varepsilon - \beta^*\right] = x(x'x)^{-1}x'\varepsilon \end{aligned}$$

hence

$$\begin{aligned} \left\|x(\hat{\beta}_{OLS} - \beta^*)\right\|_2^2 &= (\hat{\beta}_{OLS} - \beta^*)'x'x(\hat{\beta}_{OLS} - \beta^*) = \varepsilon'x(x'x)^{-1}x'\varepsilon \\ &= \mathrm{Tr}\left[\varepsilon'x(x'x)^{-1}x'\varepsilon\right] = \mathrm{Tr}\left[(x'x)^{-1}x'\varepsilon\varepsilon'x\right] \end{aligned}$$

and finally

$$\mathbb{E}\left(\left\|x(\hat{\beta}_{OLS} - \beta^*)\right\|_2^2\right) = \mathrm{Tr}\left[(x'x)^{-1}x'\mathbb{E}\left(\varepsilon\varepsilon'\right)x\right] = \sigma^2\mathrm{Tr}\left[(x'x)^{-1}x'x\right] = \sigma^2 p.$$

□

It is interesting to remark that this result is very easy to prove in the deterministic-design framework. On the other hand, establishing the same kind of result in the setting of a random-design model is much more difficult and often requires making restrictive assumptions on the distribution of X and on the vector ε. We refer, for example, to [GKKW04]. Attempting to remove these assumptions is still the subject of active research; see, for example, [AC09].

2.2 Ideas for the case $p > n$

One therefore remarks an important fact: not only is $p \leq n$ necessary in order to have uniqueness of the estimator, but the quality of the estimation is all the worse as p is large.

Yet, in a certain number of current applications, this setting is not at all realistic. One has, for example, a certain number of problems in genomics in which one observes the genome of a group of individuals suffering from a rather rare type of disease: one typically has $n = 30$ individuals. One stores information on each gene in the vector X_i of individual i, which means that p is of the order of 30000.

In this case, without additional assumptions, it is not really possible to estimate the parameter β^* correctly. Nevertheless, in this precise example, one can make the following remark: it is

plausible that only one gene, or a small group of genes, having undergone a mutation, is responsible for the disease under consideration. Only, this group of genes, and in particular their location, is unknown. One is therefore led to treat the following situation: β^* is a vector of large size p compared with n, but most of its components are zero, without knowing which ones. In other words, $p >> n$ but $n \geq |\{i \in 1, \ldots, n : \beta_i^* \neq 0\}|$. This kind of problem is called a *high-dimensional problem*, and the assumption made on β^* is a structural assumption known as the *sparsity* of the parameter β^* (in French, *parcimonie*, often gallicized as *sparsité*).

The sparsity assumption, and others, make the estimation of β^* possible. We shall see this later. If one knew in advance the set of nonzero coordinates of β^*, that is, the support of β^*: $S^* := \{i : \beta_i \neq 0\} \subset \{1, ..., p\}$, it would suffice to compute the least-squares estimator restricted to the coordinates in S^*:

$$\hat{\beta}^{(S^*)} = \arg\min \{r(\beta), \beta \in \mathbb{R}^p \text{ and } j \notin S^* \Rightarrow \beta_j = 0\}$$

and one would then have, by Theorem 2.1:

$$\mathbb{E}\left(\left\|x(\hat{\beta}^{(S^*)} - \beta^*)\right\|_2^2\right) = \sigma^2 |S^*| \ll \sigma^2 p. \tag{4}$$

Unfortunately, one does not know the set S^* (in other words, and to return to the previous example, one does not know in advance how to locate the genes responsible for the disease under consideration).

2.3 The ℓ_0 penalization

2.3.1 Definition

A first idea consists in proposing the following estimator:

$$\hat{\beta}_\lambda = \arg\min \left\{ \|y - x\beta\|_2^2 + \lambda \|\beta\|_0, \beta \in \mathbb{R}^p \right\},$$

with a $\lambda > 0$. In other words, one "penalizes" the vectors β that are not sparse enough.

This estimator was introduced in [Aka73] with $\lambda = 2\sigma^2$ under the name AIC (on the basis of an idea of [Mal73]), then in [Sch78] under the name BIC with $\lambda = \sigma^2 \log(n)$ (note that this implies that σ^2 must be known).

This estimator can be generalized in the following way:

$$\hat{\beta}_f = \arg\min \left\{ \|y - x\beta\|_2^2 + f(\|\beta\|_0), \beta \in \mathbb{R}^p \right\}$$

where $f(\cdot)$ is an arbitrary increasing function.

2.3.2 A theoretical result

Theorem 2.2 ([BTW07]) *If the X_i are deterministic, $Y_i = X_i'\beta + \varepsilon_i$ and the ε_i are i.i.d. $\mathcal{N}(0, \sigma^2)$, if one chooses*

$$f(k) = \frac{2\sigma^2 k}{n}\left(1 + \frac{3}{\sqrt{2}}\sqrt{2\log\frac{ep}{k \vee 1}} + 4\log\frac{ep}{k \vee 1}\right)$$

then

$$\mathbb{E}\left(\left\|x(\hat{\beta}_f - \beta)\right\|_2^2\right) \leq 5\|\beta\|_0\sigma^2\left(1 + \log\frac{ep}{\|\beta\|_0 \vee 1}\right) + 7\sigma^2.$$

This theorem is stated and proved in the article [BTW07]. We take it as given here. Moreover, in [BTW07], it is also shown that this bound cannot be improved (except for the constant); roughly speaking, the order of magnitude $\|\beta\|_0 \log(p/\|\beta\|_0)$ is optimal. In other words, the "price to pay" for not knowing the location of the $\|\beta\|_0$ nonzero coordinates is of the order of $\log(p/\|\beta\|_0)$.

2.3.3 The implementation problem

The estimator $\hat{\beta}_f$ therefore seems satisfactory from a theoretical point of view. Unfortunately, from a practical point of view, the problem is far from settled. Indeed, in order to compute $\hat{\beta}_f$ or even $\hat{\beta}_\lambda$ in practice, one does not know of an algorithm really faster than the one that consists in running through all possible sets of nonzero coordinates for β: for every $J \subset \{1, ..., p\}$ one computes $\hat{\beta}^{(J)}$, then one compares the values $\|y - x\hat{\beta}^{(J)}\|_2^2 + \lambda\|\hat{\beta}^{(J)}\|_0$. This implies that one must compute as many estimators as there are subsets $J \subset \{1, ..., p\}$, i.e. 2^p. For $p = 10$, this poses no problem. But in the genomics example given above, $2^p = 2^{30000} > 10^{9030}$, and this estimator is not usable in practice.

2.4 The ℓ_1 penalization

2.4.1 Definition

A natural idea is to replace the norm $\|b\|_0$ in the penalization by the norm $\|b\|_1$, which is a convex function of b: one then remains within convex optimization problems, which are much easier to handle.

One defines:

$$\tilde{\beta}_\lambda = \arg\min \left\{ \|y - x\beta\|_2^2 + \lambda\|\beta\|_1, \beta \in \mathbb{R}^p \right\}$$

for a $\lambda > 0$ to be specified. The estimator obtained bears the name LASSO (Least Angle Shrinkage and Selection Operator), introduced in [Tib96].

We shall now see that it is possible to compute this estimator very quickly, even for very large values of p. Moreover, there is a theoretical result fairly similar to Theorem 2.2. Unfortunately, the price to pay is an assumption on the design $X_1, \ldots, X_n$, which is not always satisfied.

2.4.2 Convexification of the ℓ_0 norm

The choice of the ℓ_0 function as a penalty is natural in the framework of high-dimensional statistics. As recalled in Subsection 2.3.3, this function is not convex and therefore cannot be used in practice. One is therefore led to seek the convex function closest to the ℓ_0 function. This type of function is called the *convex envelope*, whose definition we now recall.

Definition 2.3 *([HUL93]) Let $d \geq 1$ be an integer and let C be a convex subset of $\mathbb{R}^d$. Let $f : C \to \mathbb{R}$ be a function. The convex envelope of f is the largest convex function g such that $g(x) \leq f(x)$ for every $x \in C$. One denotes by* $\mathrm{conv}(f)$ *the convex envelope of f.*

This definition is meaningful, because if $\mathcal{C}$ denotes the set of all convex functions smaller than f, then $\sup_{g \in \mathcal{C}} g$ is always a convex function and is therefore equal to $\mathrm{conv}(f)$.

Theorem 2.4 *The ℓ_1 norm is the convex envelope of the ℓ_0 function on B_∞^d, the unit ball of $(\mathbb{R}^d, \ell_\infty^d)$.*

Proof: We prove the result only on B_1^d. For every $x \in B_1^d$, one has $\|x\|_1 \leq \|x\|_0$, so if F is the convex envelope of ℓ_0 on B_∞^d, one will have: $\|x\|_1 \leq F(x)$ for every $x \in B_2^d$. In particular, if one denotes by $(e_1, \ldots, e_d)$ the canonical basis of $\mathbb{R}^d$, one sees that for every $i = 1, \ldots, d$, $1 = \|\pm e_i\|_1 \leq F(\pm e_i) \leq \|\pm e_i\|_0 = 1$, hence $F(\pm e_i) = 1$. Let $\lambda = (\lambda_1, \ldots, \lambda_d)' \in \mathbb{R}^d$ be such that $\sum_{i=1}^d |\lambda_i| \leq 1$. One has, by convexity of F and since $F(0) = 0$ (because $\|0\|_1 \leq F(0) \leq \|0\|_0$),

$$\|\lambda\|_1 = \left\| \sum_{i=1}^d |\lambda_i| \mathrm{sign}(\lambda_i) e_i \right\|_1 \leq F\Big(\sum_{i=1}^d |\lambda_i| \mathrm{sign}(\lambda_i) e_i + (1 - \|\lambda\|_1) 0 \Big)$$

$$\leq \sum_{i=1}^d |\lambda_i| F(\mathrm{sign}(\lambda_i) e_i) + (1 - \|\lambda\|_1) F(0) = \sum_{i=1}^d |\lambda_i| = \|\lambda\|_1 . \tag{5}$$

Hence for every $\lambda \in B_1^d$ (the unit ball of $(\mathbb{R}^d, \ell_1^d)$), $F(\lambda) = \|\lambda\|_1$. This proves that the convex envelope of ℓ_0 on B_1^d is indeed ℓ_1. To obtain the result on B_∞^d, one appeals to the convex biconjugate of ℓ_0. We refer the interested reader to [HUL93]. □

2.4.3 A first theoretical result

We consider the regression problem with deterministic design introduced in Subsection 1.2. Let $X_1, \ldots, X_n$ be n (nonrandom) vectors in $\mathbb{R}^p$. One observes n noisy projections of a vector $\beta^* \in \mathbb{R}^p$ in the directions $X_1, \ldots, X_n$:

$$Y_i = \langle X_i, \beta^* \rangle + \epsilon_i$$

where for every $i = 1, \ldots, n$, ϵ_i is a centered random variable. We write

$$y := \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix} \in \mathbb{R}^n, \; x := \begin{pmatrix} X_1' \\ \vdots \\ X_n' \end{pmatrix} \in \mathcal{M}_{n,p} \text{ and } \vec{\epsilon} := \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

The analysis of the statistical performance of the LASSO relies strongly on the properties of the matrix x. All the assumptions on x encountered in the literature are in a certain sense of the same nature as the **(RIP) condition**, *Restricted Isometry Property*: one says that a matrix $A : \mathbb{R}^p \to \mathbb{R}^n$ satisfies the condition $(RIP)(s, \delta)$ for some $1 \leq s \leq n$ and $0 < \delta < 1$ when, for every vector $\beta \in \mathbb{R}^p$ such that $|\{i \in \{1, \ldots, p\} : \beta_i \neq 0\}| \leq s$, one has

$$(1 - \delta) \|\beta\|_2 \leq \|A\beta\|_2 \leq (1 + \delta) \|\beta\|_2 . \tag{6}$$

The relation (6) means that A behaves like a $(1 + \delta)$-isometry on the set of *s-sparse vectors*, denoted

$$U_s := \{\beta \in \mathbb{R}^p : |\mathrm{supp}(\beta)| \leq s\} \text{ where } \mathrm{supp}(\beta) = \{i \in \{1, \ldots, p\} : \beta_i \neq 0\}.$$

One can also understand this property by saying that A satisfies (RIP) when every submatrix of A of size $s \times n$ is a $(1 + \delta)$-isometry from ℓ_2^s into ℓ_2^n.

It is this kind of property that one hopes to see realized by x in compressed sensing theory. Here, unlike compressed sensing, one wishes to reconstruct/estimate only a single signal β^* and not all s-sparse signals (as is the case in compressed sensing). One will therefore have conditions bearing only on β^* (and in particular on its support). Also, in general it is the lower bound in (6) that poses the most difficulty. This type of inequality concerns "the smallest singular value of A

restricted to subsets of $\mathbb{R}^p$" (the expression is in quotation marks because, when A is restricted to a subset of $\mathbb{R}^p$ that is not a linear space, it does not really make sense to speak of singular values).

Another example of a property of x is given by the condition $\Phi(L, \mathrm{supp}(\beta^*)) > 0$, where for every $L > 0$ and $S \subset \{1, \dots, p\}$

$$\Phi(L, S) := \min\Big(\frac{|S|}{n} \|x\beta\|_2^2 : \|\beta_S\|_1 = 1, \|\beta_{S^c}\|_1 \leq L \Big) \tag{7}$$

where for every $\beta \in \mathbb{R}^p$, one denotes by β_S the vector in $\mathbb{R}^p$ such that $(\beta_S)_i = 0$ when $i \notin S$ and $(\beta_S)_i = \beta_i$ when $i \in S$.

Theorem 2.5 *We write $S^* := \mathrm{supp}(\beta^*)$. We consider the event*

$$\Omega_0 := \Big\{ 2 \left\| x' \vec{\epsilon} \right\|_\infty \leq \lambda_0 \Big\}.$$

We take $\lambda > \lambda_0$ and write $L = (\lambda + \lambda_0)/(\lambda - \lambda_0)$. We assume that $\Phi(L, S^*) > 0$. Then, on the event Ω_0, one has

1. $$\left\| x\tilde{\beta}_\lambda - x\beta^* \right\|_2^2 \leq \frac{4\lambda^2 |S^*|}{n\Phi(L, S^*)};$$

2. $$\left\| \tilde{\beta}_\lambda - \beta^* \right\|_1 \leq \frac{4\lambda^2 |S^*|}{n(\lambda - \lambda_0)\Phi(L, S^*)}.$$

Proof: A classical ingredient in this kind of proof is the *dual bound*:

$$|\langle \vec{\epsilon}, x\tilde{\beta}_\lambda - x\beta^* \rangle| \leq \left\| x'\vec{\epsilon} \right\|_\infty \left\| \tilde{\beta}_\lambda - \beta^* \right\|_1.$$

In particular, on the event Ω_0, one has $|2\langle \vec{\epsilon}, x\tilde{\beta}_\lambda - x\beta^* \rangle| \leq \lambda_0 \left\| \tilde{\beta}_\lambda - \beta^* \right\|_1$.

One then has, by definition of $\tilde{\beta}_\lambda$,

$$\left\| y - x\tilde{\beta}_\lambda \right\|_2^2 + \lambda \left\| \tilde{\beta}_\lambda \right\|_1 \leq \|y - x\beta^*\|_2^2 + \lambda \|\beta^*\|_1.$$

Hence on Ω_0, one will have

$$\begin{aligned}
&\left\| x\tilde{\beta}_\lambda - x\beta^* \right\|_2^2 \leq 2\langle \vec{\epsilon}, x\beta^* - x\tilde{\beta}_\lambda \rangle + \lambda\big(\|\beta^*\|_1 - \left\| \tilde{\beta}_\lambda \right\|_1 \big) \\
&\leq \lambda_0 \left\| \tilde{\beta}_\lambda - \beta^* \right\|_1 + \lambda\big(\|\beta^*\|_1 - \left\| \tilde{\beta}_\lambda \right\|_1 \big) \\
&= \lambda_0 \left\| (\tilde{\beta}_\lambda)_{S^*} - \beta^* \right\|_1 + \lambda_0 \left\| (\tilde{\beta}_\lambda)_{(S^*)^c} \right\| + \lambda\big(\|\beta^*\|_1 - \left\| (\tilde{\beta}_\lambda)_{S^*} \right\|_1 \big) - \lambda \left\| (\tilde{\beta}_\lambda)_{(S^*)^c} \right\|_1 \\
&\leq (\lambda_0 - \lambda) \left\| (\tilde{\beta}_\lambda)_{(S^*)^c} \right\|_1 + (\lambda_0 + \lambda) \left\| (\tilde{\beta}_\lambda)_{S^*} - \beta^* \right\|_1.
\end{aligned}$$

Since $\lambda > \lambda_0$, one deduces that, on the event Ω_0,

$$\left\| x\tilde{\beta}_\lambda - x\beta^* \right\|_2^2 \leq 2\lambda \left\| (\tilde{\beta}_\lambda)_{S^*} - \beta^* \right\|_1 \text{ and } \left\| (\tilde{\beta}_\lambda)_{(S^*)^c} \right\|_1 \leq \frac{\lambda_0 + \lambda}{\lambda - \lambda_0} \left\| (\tilde{\beta}_\lambda)_{S^*} - \beta^* \right\|_1. \tag{8}$$

Moreover, it follows from the assumption $\Phi(L, S^*) > 0$ applied to the vector $\nu := \big(\tilde{\beta}_\lambda - \beta^*\big) / \Big\|(\tilde{\beta}_\lambda)_{S^*} - \beta^*\Big\|_1$, which is such that $\|\nu_{S^*}\|_1 = 1$ and $\big\|\nu_{(S^*)^c}\big\|_1 = \Big\|(\tilde{\beta}_\lambda)_{(S^*)^c}\Big\|_1 / \Big\|(\tilde{\beta}_\lambda)_{S^*} - \beta^*\Big\|_1 \leq \frac{\lambda_0+\lambda}{\lambda-\lambda_0} = L$, that

$$\frac{|S^*|}{n}\Big\|x\big(\tilde{\beta}_\lambda - \beta^*\big)\Big\|_2^2 \geq \Phi(L, S^*)\Big\|(\tilde{\beta}_\lambda)_{S^*} - \beta^*\Big\|_1^2.$$

It then follows in (8) that

$$\Big\|x\tilde{\beta}_\lambda - x\beta^*\Big\|_2^2 \leq 2\lambda\sqrt{\frac{|S^*|}{n\Phi(L, S^*)}}\Big\|x\big(\tilde{\beta}_\lambda - \beta^*\big)\Big\|_2$$

whence point *1)* follows, and for point *2)* it suffices to observe that, thanks to (8),

$$\begin{aligned}
&\Big\|\tilde{\beta}_\lambda - \beta^*\Big\|_1 = \Big\|(\tilde{\beta}_\lambda)_{(S^*)^c}\Big\|_1 + \Big\|(\tilde{\beta}_\lambda)_{S^*} - \beta^*\Big\|_1 \\
&\leq (1+L)\Big\|(\tilde{\beta}_\lambda)_{S^*} - \beta^*\Big\|_1 \leq \frac{2\lambda}{\lambda-\lambda_0}\sqrt{\frac{|S^*|}{n\Phi(L, S^*)}}\Big\|x\big(\tilde{\beta}_\lambda - \beta^*\big)\Big\|_2 \\
&\leq \frac{4\lambda^2}{\lambda-\lambda_0}\frac{|S^*|}{n\Phi(L, S^*)}.
\end{aligned}$$

□

By the previous theorem, one must choose the regularization parameter λ larger than λ_0, where λ_0 is a bound on the noise such that $2\|x'\vec{\epsilon}\|_\infty \leq \lambda_0$. For the result of Theorem 2.5 to be meaningful, one must ensure that $2\|x'\vec{\epsilon}\|_\infty \leq \lambda_0$ holds with high probability.

To obtain an upper bound for $\|x'\vec{\epsilon}\|_\infty$ that holds with high probability, one uses the following argument: for every $t > 0$

$$\begin{aligned}
&\mathbb{P}\big[\,\big\|x'\vec{\epsilon}\big\|_\infty \geq t\big] = \mathbb{P}\big[\exp\big(\lambda\big\|x'\vec{\epsilon}\big\|_\infty\big) \geq \exp(\lambda t)\big] \leq \exp(-\lambda t)\mathbb{E}\exp\big(\lambda\big\|x'\vec{\epsilon}\big\|_\infty\big) \\
&\leq \exp(-\lambda t)\sum_{i=1}^{p}\mathbb{E}\exp\big(\lambda\big\langle C_j, \vec{\epsilon}\big\rangle\big),
\end{aligned}$$

where $\lambda > 0$ is a parameter to be optimized afterwards and $C_1, \ldots, C_p$ are the column vectors of x. If $\epsilon_1, \ldots, \epsilon_n$ are independent, then for every vector $v \in \mathbb{R}^n$, one has

$$\mathbb{E}\exp\big(\lambda\big\langle v, \vec{\epsilon}\big\rangle\big) = \mathbb{E}\prod_{i=1}^{n}\exp(\lambda v_i\epsilon_i) = \prod_{i=1}^{n}\mathbb{E}\exp(\lambda v_i\epsilon_i).$$

One is therefore led to control quantities of the type $\mathbb{E}\exp(\mu\epsilon)$. The function $\mu \to \mathbb{E}\exp(\mu\epsilon)$ is called the Laplace transform of ϵ. There are numerous results on this transform. We recall here an upper bound on the Laplace transform of a Gaussian random variable and of a bounded random variable (Hoeffding's inequality).

Proposition 2.6 *Let g be a Gaussian random variable $\mathcal{N}(m, \sigma)$. One has, for every $\mu > 0$,*

$$\mathbb{E}\exp\big(\mu g\big) = \exp\Big(\frac{\mu^2\sigma^2}{2} + m\mu\Big).$$

Let ϵ be a random variable such that $a \leq \epsilon \leq b$ a.s. Then, for every $\mu > 0$, one has

$$\mathbb{E}\exp\big(\mu\epsilon\big) \leq \exp\Big(\frac{\mu^2(a-b)^2}{8}\Big).$$

It is interesting to remark that in both cases the exponent of μ in the exponential is 2. This type of random variable is called a *sub-Gaussian* random variable. In particular, bounded random variables are sub-Gaussian. This observation is due to Hoeffding.

To simplify the end of the argument, we assume that $\epsilon_1, \dots, \epsilon_n$ are i.i.d. $\mathcal{N}(0, \sigma)$. One therefore has

$$\mathbb{E} \exp\big(\lambda \langle C_j, \vec{\epsilon} \rangle\big) = \prod_{i=1}^{n} \mathbb{E} \exp(\lambda C_{ji} \epsilon_i) = \prod_{i=1}^{n} \exp\Big(\frac{\lambda^2 C_{ji}^2 \sigma^2}{2}\Big) = \exp\Big(\frac{\lambda^2 \sigma^2}{2} \|C_j\|_2^2\Big)$$

and, in view of the choice of renormalization of x, one has $\|C_j\|_2^2 = n$. One therefore deduces that

$$\mathbb{P}\big[\, \big\|x'\vec{\epsilon}\big\|_\infty \geq t\big] \leq p \inf_{\lambda > 0} \exp\big(-\lambda t + n\lambda^2\sigma^2/2\big) = p \exp\big(-t^2/(2n\sigma^2)\big).$$

We fix $0 < \eta < 1$. Then, for $t = \sigma\big(2n \log(p/\eta)\big)^{1/2}$, one obtains, with probability greater than $1 - \eta$,

$$\big\|x'\vec{\epsilon}\big\|_\infty \leq \sigma\big(2n \log(p/\eta)\big)^{1/2}.$$

One deduces from Theorem 2.5 and from the previous deviation calculation the following corollary.

Corollary 2.7 *We assume that the noise terms $\epsilon_1, \dots, \epsilon_n$ are i.i.d. $\mathcal{N}(0, \sigma^2)$. Let $0 < \eta < 1$. We choose as regularization parameter $\lambda = 4\sigma\big(2n \log(p/\eta)\big)^{1/2}$, that is $\lambda = 2\lambda_0$ with $\lambda_0 = 2\sigma\big(2n \log(p/\eta)\big)^{1/2}$, so that $L = 3$. We assume that $\Phi(3, S^*) > 0$. One has, with probability greater than $1 - \eta$,*

1. $$\Big\|x\tilde{\beta}_\lambda - x\beta^*\Big\|_2^2 \leq \frac{128\sigma^2 |S^*| \log(p/\eta)}{\Phi(3, S^*)};$$

2. $$\Big\|\tilde{\beta}_\lambda - \beta^*\Big\|_1 \leq \frac{32\sqrt{2}\,\sigma |S^*|}{\Phi(3, S^*)} \sqrt{\frac{\log(p/\eta)}{n}}.$$

In particular, when $\Phi(3, S^*) \geq c_0 > 0$ where c_0 is an absolute constant, one remarks that one obtains almost the same bound as the one obtained in (4) when one knows the support of β^*. There is therefore only a loss of order $\log p$. This means that the LASSO estimator is particularly well suited to variable-selection problems in high dimension (here p may be much larger than n).

2.4.4 A second theoretical result

We seek here to weaken the previous condition $\Phi(L, S^*) > 0$.

Theorem 2.8 ([BRT10]) *We still assume that the X_i are deterministic, $Y_i = X_i'\beta + \varepsilon_i$ and the ε_i are i.i.d. $\mathcal{N}(0, \sigma^2)$. We assume moreover that the matrix x satisfies the following assumption:*

$$\exists c \in \mathbb{R}_+^*,\ \frac{1}{c} \leq \inf\Bigg\{ \frac{v'\left(\frac{x'x}{n}\right)v}{\sum_{j \in J} v_j^2},\ v \in \mathbb{R}^p,\ J \subset \{1, ..., p\},\ \sum_{j \notin J} |v_j| \leq 3 \sum_{j \in J} |v_j|,\ |J| \leq \|\beta\|_0 \Bigg\}. \quad (9)$$

Then, with the choice

$$\lambda = 4\sigma\sqrt{2n\log\frac{p}{\eta}}$$

one has

$$\mathbb{P}\left(\left\|x(\tilde{\beta}_\lambda - \beta)\right\|_2^2 \leq 128c\sigma^2\|\beta\|_0\log\frac{p}{\eta}\right) \geq 1-\eta.$$

Let us begin by examining the assumption given by Equation (9). If one forgets the conditions on v and J, one simply obtains that

$$\inf\left\{\frac{v'\left(\frac{x'x}{n}\right)v}{\|v\|_2^2}, v \in \mathbb{R}^p\right\} \geq \frac{1}{c} > 0$$

so this holds as soon as $1/c$ is smaller than or equal to the smallest eigenvalue of $x'x/n$. In other words, it suffices that $x'x/n$ be invertible for this assumption to be satisfied. Of course, when $p > n$, the matrix $x'x/n$ is never invertible. However, the condition

$$\inf\left\{\frac{v'\left(\frac{x'x}{n}\right)v}{\|v\|_2^2}, v \in F\right\} \geq \frac{1}{c} > 0,$$

where F is a subset of $\mathbb{R}^p$, can still be satisfied.

We shall now prove this theorem by following the proof of [BRT10]. We begin by giving a lemma and its proof.

Lemma 2.9 *One has*

$$\mathbb{P}\left(\|x'\varepsilon\|_\infty \leq \sigma\sqrt{2n\log\frac{p}{\eta}}\right) \geq 1-\eta.$$

Proof of Lemma 2.9: By the assumptions, $\varepsilon \sim \mathcal{N}(0,\sigma^2 I_n)$ and therefore $x'\varepsilon \sim \mathcal{N}(0,\sigma^2 x'x)$. Now, the matrix $x'x/n$ has only 1's on the diagonal; this means that the i-th component $(x'\varepsilon)_i \sim \mathcal{N}(0,\sigma^2 n)$. Hence, for $t \geq 0$,

$$\mathbb{P}\left(|(x'\varepsilon)_i| > t\right) \leq e^{\frac{-t^2}{2n\sigma^2}}.$$

Finally,

$$\mathbb{P}\left(\|x'\varepsilon\|_\infty > t\right) \leq \sum_{i=1}^{p}\mathbb{P}\left(|(x'\varepsilon)_i| > t\right) \leq pe^{\frac{-t^2}{2n\sigma^2}}$$

and one obtains the announced result by taking

$$t = \sigma\sqrt{2n\log\frac{p}{\eta}}.$$

□

Proof of Theorem 2.8: By definition,

$$\|y - x\tilde{\beta}_\lambda\|_2^2 + \lambda\|\tilde{\beta}_\lambda\|_1 \leq \|y - x\beta\|_2^2 + \lambda\|\beta\|_1$$

that is,

$$-2\tilde{\beta}_\lambda' x'y + \tilde{\beta}_\lambda' x'x\tilde{\beta}_\lambda + 2\beta' x'y - \beta' x'x\beta \leq \lambda\left(\|\beta\|_1 - \|\tilde{\beta}_\lambda\|_1\right).$$

Now $y = x\beta + \varepsilon$, hence

$$-2\tilde{\beta}'_\lambda x'x\beta + \tilde{\beta}'_\lambda x'x\tilde{\beta}_\lambda + 2\beta' x'x\beta - \beta' x'x\beta \leq 2(\tilde{\beta}'_\lambda - \beta')x'\varepsilon + \lambda\left(\|\beta\|_1 - \|\tilde{\beta}_\lambda\|_1\right)$$

that is,

$$-2\tilde{\beta}'_\lambda x'x\beta + \tilde{\beta}'_\lambda x'x\tilde{\beta}_\lambda + \beta' x'x\beta \leq 2(\tilde{\beta}'_\lambda - \beta')x'\varepsilon + \lambda\left(\|\beta\|_1 - \|\tilde{\beta}_\lambda\|_1\right)$$

that is, again,

$$\|x(\tilde{\beta}_\lambda - \beta)\|_2^2 \leq 2(\tilde{\beta}'_\lambda - \beta')x'\varepsilon + \lambda\left(\|\beta\|_1 - \|\tilde{\beta}_\lambda\|_1\right).$$

Hence,

$$\|x(\tilde{\beta}_\lambda - \beta)\|_2^2 \leq 2\|\tilde{\beta}_\lambda - \beta\|_1 \|x'\varepsilon\|_\infty + \lambda\left(\|\beta\|_1 - \|\tilde{\beta}_\lambda\|_1\right).$$

From now on, we place ourselves on the event $\{\omega \in \Omega : \|x'\varepsilon(\omega)\|_\infty \leq \sigma\sqrt{2n\log\frac{p}{\eta}}\}$ (by Lemma 2.9, this is an event of probability at least $1-\eta$). Now, since $\lambda = 4\sigma\sqrt{2n\log\frac{p}{\eta}}$, one has

$$\|x(\tilde{\beta}_\lambda - \beta)\|_2^2 \leq \frac{\lambda}{2}\left(\|\tilde{\beta}_\lambda - \beta\|_1\right) + \lambda\left(\|\beta\|_1 - \|\tilde{\beta}_\lambda\|_1\right)$$

and therefore

$$\begin{aligned}
\|x(\tilde{\beta}_\lambda - \beta)\|_2^2 + \frac{\lambda}{2}\left(\|\tilde{\beta}_\lambda - \beta\|_1\right) &\leq \lambda\left(\|\tilde{\beta}_\lambda - \beta\|_1 + \|\beta\|_1 - \|\tilde{\beta}_\lambda\|_1\right) \\
&= \lambda\left[\sum_{j=1}^p |(\tilde{\beta}_\lambda)_j - \beta_j| + \sum_{j=1}^p \left(|\beta_j| - |(\tilde{\beta}_\lambda)_j|\right)\right] \\
&= \lambda\left[\sum_{j:\beta_j\neq 0} |(\tilde{\beta}_\lambda)_j - \beta_j| + \sum_{j:\beta_j\neq 0} \left(|\beta_j| - |(\tilde{\beta}_\lambda)_j|\right)\right] \\
&\leq 2\lambda \sum_{j:\beta_j\neq 0} |(\tilde{\beta}_\lambda)_j - \beta_j|.
\end{aligned}$$

We highlight this last inequality, which we shall use several times:

$$\|x(\tilde{\beta}_\lambda - \beta)\|_2^2 + \frac{\lambda}{2}\left(\|\tilde{\beta}_\lambda - \beta\|_1\right) \leq 2\lambda \sum_{j:\beta_j\neq 0} |(\tilde{\beta}_\lambda)_j - \beta_j|. \tag{10}$$

In a first step, (10) implies

$$\frac{\lambda}{2}\left(\|\tilde{\beta}_\lambda - \beta\|_1\right) \leq 2\lambda \sum_{j:\beta_j\neq 0} |(\tilde{\beta}_\lambda)_j - \beta_j|$$

that is,

$$\|\tilde{\beta}_\lambda - \beta\|_1 \leq 4 \sum_{j:\beta_j\neq 0} |(\tilde{\beta}_\lambda)_j - \beta_j|$$

that is, again,

$$\sum_{j:\beta_j=0} |(\tilde{\beta}_\lambda)_j - \beta_j| \leq 3 \sum_{j:\beta_j\neq 0} |(\tilde{\beta}_\lambda)_j - \beta_j|.$$

Hence, taking $v = \tilde{\beta}_\lambda - \beta$ in Equation (9), as well as $J = \{j \in \{1, ..., p\} : \beta_j \neq 0\}$, one obtains that

$$\sum_{j:\beta_j \neq 0} [(\tilde{\beta}_\lambda)_j - \beta_j]^2 \leq \frac{c}{n}(\tilde{\beta}_\lambda - \beta)' x' x (\tilde{\beta}_\lambda - \beta) = \|x(\tilde{\beta}_\lambda - \beta)\|_2^2. \tag{11}$$

Next, we start again from Inequality (10) in order to write:

$$\begin{aligned}
\|x(\tilde{\beta}_\lambda - \beta)\|_2^2 &\leq 2\lambda \sum_{j:\beta_j \neq 0} |(\tilde{\beta}_\lambda)_j - \beta_j| \\
&\leq 2\lambda \sqrt{\|\beta\|_0 \frac{c}{n} \sum_{j:\beta_j \neq 0} [(\tilde{\beta}_\lambda)_j - \beta_j]^2} \\
&\leq 2\lambda \sqrt{\|\beta\|_0 \frac{c}{n} \|x(\tilde{\beta}_\lambda - \beta)\|_2^2}
\end{aligned}$$

by (11), and therefore:

$$\|x(\tilde{\beta}_\lambda - \beta)\|_2^2 \leq 4\lambda^2 \frac{c}{n} \|\beta\|_0.$$

□

2.4.5 Computational algorithms

First of all, note that the set of all solutions of

$$\min \left\{ \|y - xb\|_2^2 + \lambda \|b\|_1, b \in \mathbb{R}^p \right\}$$

when $\lambda \in \mathbb{R}_+$ is the same as the set of all solutions of

$$\begin{cases} \min_{b \in \mathbb{R}^p} \|y - xb\|_2^2 \\ \\ \text{under the constraint } \|b\|_1 \leq t \end{cases} \tag{12}$$

when $t \in \mathbb{R}_+^*$. Indeed, problem (12) is a problem of minimizing a convex function under a convex constraint with a constraint qualification, so one can apply the duality theorem.

A first algorithm makes it possible to approximate the solutions of this optimization problem: one fixes a (very small) threshold $\zeta > 0$ and one starts from $\beta(0) = 0$. One then updates the parameter in the following way:

$$\forall j \in \{1, ..., p\}, \quad c_j^{(k)} = x_j'(y - x\hat{\beta}(k)$$

$$j(k) = \arg \max_{j \in \{1,...,p\}} |c_j^{(k)}|,$$

in other words, one seeks the column most correlated with the current residual, then one sets

$$\beta(k+1) = \beta(k) + \zeta \mathrm{sgn}(\mathrm{c}_{\mathrm{j(k)}}^{(\mathrm{k})}) \mathbb{I}_j$$

where for every $j \in \{1, ..., p\}$, $\mathbb{I}_j = (\mathbb{1}_{i=j})'_{1 \leq i \leq p}$. Intuitively, one has (at least as long as there is no "backtracking"), $\|\beta(k)\|_1 = \zeta k$, so one has the impression of following (in a discretized way) the path of the solutions of problem (12).

We shall present here an algorithm that finds all the solutions of problem (12) for all values of $t \in \mathbb{R}_+^*$. This algorithm bears the name LARS and was introduced for the first time in [EHJT04].

One can see it as a refined version of the previous algorithm, which detects the moment at which a new variable comes into play.

The method is also iterative: it starts from $\beta(0) = 0$ and at each step k, one adds a nonzero component, so that $\|\beta(k)\|_0 = k$ (up to $k = n$). At step k, the set of nonzero coordinates of $\beta(k)$ is denoted $\mathcal{A}(k)$. For $\mathcal{A} \subset \{1, ..., p\}$ one denotes by $x_{\mathcal{A}}$ the matrix x from which one has removed all columns with index $j \notin \mathcal{A}$. Formally, if one denotes by x_1, ..., x_p the columns of x,

$$x = (x_1 \dots x_p) \text{ and } x_{\mathcal{A}} = (x_j)_{j \in \mathcal{A}}.$$

One then has:

$$\mathcal{A}(k+1) = \mathcal{A}(k) \cup \arg\max_j |x_j'(y - x\hat{\beta}(k)|.$$

One therefore simply seeks the column x_j most correlated with the current residual $y - x\hat{\beta}(k)$. One may still write $c_j^{(k)} = x_j'(y - x\hat{\beta}(k)$ in what follows. It then remains to determine how to update the coordinates with index $j \in \mathcal{A}(k+1)$. In fact, one has:

$$\beta(k+1) = \beta(k) + \gamma(k+1)w(k+1)$$

where

$$\gamma_{k+1} = \min\left(\frac{c - c_j}{1 - a_j}, \frac{c + c_j}{1 + a_j}\right)$$

with

$$c = \|x'_{\mathcal{A}(k+1)}(y - x\beta(k))\|_\infty,$$

$$a_j = x_j' w_{k+1}$$

and finally $w_{k+1} \in \mathbb{R}^p$ is such that, if the u_j, $j \in \mathcal{A}(k+1)$, are a solution of

$$x'_{\mathcal{A}(k+1)} x_{\mathcal{A}(k+1)} (u_j)_{j \in \mathcal{A}(k+1)} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

then

$$w_{k+1} = \sum_{j \in \mathcal{A}(k+1)} u_j \mathbb{I}_j.$$

Proposition 2.10 *The set of solutions of problem* (12) *is a piecewise linear path whose nodes are the points* $\beta(0)$, $\beta(1)$, ...

We take it as given here; see, for example, the original article [EHJT04].

In practice, there is no need to implement this algorithm; this is very well done in a whole range of software packages (MatLab, SciLab, R). For example, in R, the instruction lars(y, x) computes all the solutions and plot(lars(y, x)) represents them graphically ("regularization path").

2.4.6 Link with compressed sensing

One can establish a link with another mathematical problem, very much in fashion at present in signal processing, that of *compressed sensing*. This problem is defined formally in the following way: one observes $y \in \mathbb{R}^n$ satisfying $y = x\beta$ with $\beta \in \mathbb{R}^p$ and x a known $n \times p$ matrix. Of course,

if $p > n$, the matrix x is not invertible, and recovering the vector β is impossible, because the system $y = xb$ admits infinitely many solutions. However, the object

$$\beta^* = \arg\min_b \{\|b\|_0 : y = xb\}$$

can be defined in a unique way and is of particular interest in signal processing. But for the moment there is no known algorithm allowing one to compute it efficiently.

Specialists of the problem then turned their attention to:

$$\tilde{\beta} = \arg\min_b \{\|b\|_1 : y = xb\}.$$

This object can be computed efficiently. The question is then: does one have $\tilde{\beta} = \beta^*$? Theorems established recently show that, under conditions extremely similar to (9) on the matrix x, the answer is yes.

2.5 The ℓ_2 penalization

2.5.1 Motivation: Tikhonov regularization

The following problems, called "inverse problems", are very often studied in statistics, signal processing, and so on. Let a signal s (a vector, a function, ...) and an operator A that associates with s the object As, itself a vector, a function, and so on.

An inverse problem is the problem of reconstructing s from the datum $D = As$. Roughly speaking, if the operator A is invertible, this does not pose too many problems. However, several difficulties may arise:

- it may be that the operator A is not invertible, or that its inverse is difficult to compute;
- it may be that the observation is noisy: one does not observe $D = As$ but $D = As+\xi$, where ξ is a noise (modeled in statistics by a random variable, but sometimes by a deterministic object in signal processing).

In the second case, it is frequent that even if A is invertible, its inverse is rather unstable: $A^{-1}D = A^{-1}(As + \xi)$ may be very different from s even with ξ small. Think of the trivial case in which $s \in \mathbb{R}^2$ and

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 10^{-5} \end{pmatrix}.$$

The idea of Tikhonov regularization is to consider the solution of the minimization problem:

$$\min_t \left[\|At - D\| + \lambda\|t\|_2^2\right]$$

(norms to be specified according to the objects under consideration). Note that $t = A^{-1}D$ is a solution if $\lambda = 0$; otherwise, the solution is different.

2.5.2 Ridge estimator

Applying Tikhonov's idea to the linear regression problem, where one indeed has $y = x\beta + \varepsilon$, so that here the datum is $D = y$, the noise is $\xi = \varepsilon$, the object is $s = \beta$ and the operator A is the matrix x, one obtains the following definition:

$$\beta_\lambda^* = \arg\min \left\{ \|y - xb\|_2^2 + \lambda\|b\|_2^2, b \in \mathbb{R}^p \right\}.$$

This estimator was studied for the first time in statistics in the article [HK70] under the name "ridge regression". In fact, unlike the LASSO, one can compute this estimator explicitly.

Theorem 2.11 *If $\lambda > 0$ one has $\beta_\lambda^* = (x'x + \lambda I_p)^{-1} x'y$.*

Proof: One differentiates

$$F(b) = \|y - xb\|_2^2 + \lambda \|b\|_2^2 = (y - xb)'(y - xb) + \lambda b'b$$

and one obtains

$$\nabla F(b) = -2y'xb + 2x'xb + 2\lambda b$$

which vanishes at $(x'x + \lambda I_p)^{-1} x'y$; moreover the Hessian is $2(x'x + I_p)$, which is symmetric positive definite.
□

2.5.3 Discussion

In fact, the ridge estimator is not particularly useful in the case of sparsity. It is, however, very often used in practice. Note that:

- for a given $\lambda > 0$, this estimator is always defined (unlike the least-squares estimator, which poses a problem if $x'x$ is not invertible, in particular when $p > n$);
- it may be that $x'x$ is invertible but that its smallest eigenvalue, $\lambda_{\min}(x'x)$, is very small. In that case, the inversion of $x'x$ may be numerically unstable, which will not be the case for that of $x'x + \lambda I_p$ if λ is large enough;
- in particular, this is the case if there are very strongly correlated variables in the regression (in other words, columns of the matrix x that are almost identical).

One can find a very good discussion of the pleasant practical properties of β_λ^* in [HTF09] (in general, this book is recommended for its very good intuitions on a whole range of methods in statistical learning, although it does not present the theory).

3 Aggregation methods

We present here a few aggregation methods in the learning model of Subsection 1.1. That is, one has n observations $(X_1, Y_1), \ldots, (X_n, Y_n)$ that are i.i.d. random variables with the same distribution as (X, Y). In the general framework of statistical learning, the output Y takes values in $\mathbb{R}$ and the input X takes its values in an arbitrary abstract space $\mathcal{X}$. In classification, Y is assumed to take its values in $\{0, 1\}$. Here, we are interested in the prediction of Y as a function of X. That is, one seeks to construct, from the data $\mathcal{D}$, a function $\hat{f}$ such that Y is well estimated by $\hat{f}(X)$. In Subsection 1.1, the emphasis was placed on linear estimators, that is, estimators of the form $\hat{f}(X) = \langle X, \hat{\beta} \rangle$. Here we do not restrict ourselves only to this type of estimator.

In this chapter, we shall use the quadratic risk to present a few aggregation methods. The quadratic risk of a function $f : \mathcal{X} \to \mathbb{R}$ is given by

$$R(f) = \mathbb{E}\big(Y - f(X)\big)^2.$$

This risk measures the quality of the prediction of Y by $f(X)$. It cannot be computed from the observations $\mathcal{D}$ alone. An empirical version (i.e. computable from the data) is given by the quadratic empirical risk

$$r(f) := \frac{1}{n} \sum_{i=1}^{n} (Y_i - f(X_i))^2.$$

All the aggregation methods that we introduce in what follows are valid for a more general risk function of the type $R(f) = \mathbb{E}\ell((X,Y),f)$ and an empirical risk $R_n(f) = n^{-1}\sum_{i=1}^n \ell((X_i,Y_i),f)$, where ℓ is called the *loss function*. In particular, in the classification model, one may consider the loss function $\ell((X,Y),f) = \mathbb{1}_{[f(X)\neq Y]}$ for every classification function f with values in $\{0,1\}$.

3.1 Principle of aggregation methods

There exists a multitude of ways of constructing predictors of the output Y from the data set $\mathcal{D}$:

1. Estimators by projection onto different bases and, in general, the least-squares estimator on a class of functions F: $\hat{f} \in \arg\min_{f\in F} r(f)$;

2. Estimators by local polynomials:

$$\hat{f}(x) \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^n (Y_i - a)^2 K\Big(\frac{X_i - x}{h}\Big)$$

   where $K : \mathbb{R} \to \mathbb{R}^+$ is called a *kernel* and $h > 0$ a window.

3. Weighted estimators:

$$\hat{f}(x) = \sum_{i=1}^n W_i(x) Y_i$$

   where $W_1, \ldots, W_n$ are weight functions. For example, $W_i(x) = k^{-1} I(X_i \in J_k(x)$ where $J_k(x)$ is the set of the k observations closest to x for some $k = 1, \ldots, n$. In this case $\hat{f}$ is called the *k-nearest neighbors* estimator. Or else

$$W_i(x) = \frac{K\big((X_i - x)/h\big)}{\sum_{j=1}^n K\big((X_j - x)/h\big)}$$

   for a kernel K and a window $h > 0$. This estimator is called the Nadaraya–Watson estimator.

It is sometimes difficult to determine which prediction method will be more efficient merely by looking at the data $(X_1,Y_1),\ldots,(X_n,Y_n)$. It is also sometimes necessary to choose carefully a parameter of the estimator: the window h or the number k of nearest neighbors. This kind of problem is called an adaptation problem, or an estimator selection problem. To solve this kind of problem, one may appeal to aggregation methods. We shall call an aggregation method in the broad sense any method used to choose a particular estimator or to determine a parameter.

3.2 Presentation of the problem and partition of the data

The aggregation problem is posed as follows: given a family of M estimators $(\hat{f}_1, \ldots \hat{f}_M)$, one wishes to construct a new estimator $\tilde{f}$ such that the risk of this estimator is of the same order of magnitude as the risk of the best element in $\{\hat{f}_1, \ldots \hat{f}_M\}$. One therefore seeks to prove results of the type: "With high probability,

$$R(\tilde{f}) \leq C \min_{1\leq j\leq M} R(\hat{f}) + r_n(M) \tag{13}$$

where $C \geq 1$ and $r_n(M)$ is called the remainder." Inequalities of the type (13) are called *oracle inequalities* because an element $\hat{f} \in \{\hat{f}_1, \ldots \hat{f}_M\}$ such that $R(\hat{f}) = \min_{1\leq j\leq M} R(\hat{f})$ is called an *oracle estimator*. One may also seek to establish oracle inequalities in expectation:

$$\mathbb{E}R(\tilde{f}) \leq C \min_{1\leq j\leq M} \mathbb{E}R(\hat{f}) + r_n(M). \tag{14}$$

Aggregation methods are based on a simple principle. Rather than constructing our estimators $\hat{f}_1, \ldots, \hat{f}_M$ from all of our data, one will use only part of these data to construct these estimators, and the rest of these data will be used as a test sample. That is, part of the data serves to construct estimators and the rest to test them. Formally, one is given two integers n_T and n_V such that $n_T + n_V = n$. One considers the sample

$$\mathcal{D}_T := \{(X_1, Y_1), \ldots, (X_{n_T}, Y_{n_T})\}$$

called the *learning sample* (*training set*). This sample will serve us to construct a family of estimators $\hat{f}_1, \ldots, \hat{f}_M$; in fact $\hat{f}_1 = \hat{f}_1(\mathcal{D}_T), \ldots, \hat{f}_M = \hat{f}_M(\mathcal{D}_T)$. This first step in the construction of aggregation methods is called the *training phase*.

The second phase is the *validation phase*, or *learning phase*. The rest of the data forms the *test sample*, or *validation sample* (*test set*); it is defined by

$$\mathcal{D}_V := \{(X_{n_T+1}, Y_{n_T+1}), \ldots, (X_n, Y_n)\}.$$

This sample will serve us to test the estimators $\hat{f}_1(\mathcal{D}_T), \ldots, \hat{f}_M(\mathcal{D}_T)$.

The first aggregation method that may come to mind is to determine the estimator minimizing the empirical risk on $\mathcal{D}_V$ defined by

$$r_V(f) = \frac{1}{n_V} \sum_{i=n_T+1}^{n} (Y_i - f(X_i))^2.$$

Concretely, one will choose

$$\tilde{f} = \tilde{f}(\mathcal{D}) \in \arg \min_{f \in \{\hat{f}_1(\mathcal{D}_T), \ldots, \hat{f}_M(\mathcal{D}_T)\}} r_V(f).$$

This method is called the *empirical risk minimization method*. Most aggregation methods are derived from this basic method.

3.3 Cross-validation method

This method has the advantage of using, in a certain sense, all the data for both the learning and the validation phases. The cross-validation method is also known under the name "leave-one-out", because one constructs the estimators with $n-1$ observations and the last observation is used as a test.

For a given estimator $\hat{f}$, one constructs the cross-validation risk (denoted VC in what follows) by

$$r_{VC}(\hat{f}) := \frac{1}{n} \sum_{i=1}^{n} \left(Y_i - \hat{f}(\mathcal{D}_i)(X_i)\right)^2$$

where for every $i = 1, \ldots, n$, $\mathcal{D}_i := \{(X_j, Y_j) : j \neq i\}$ and $\hat{f}(\mathcal{D}_i)$ is the estimator constructed from the $n-1$ observations of $\mathcal{D}_i$.

The cross-validation method is then given by

$$\tilde{f} \in \arg \min_{f \in \{\hat{f}_1, \ldots, \hat{f}_M\}} r_{VC}(f).$$

For example, imagine that we are working with the k-nearest neighbors method and that we wish to choose the parameter k empirically. For every $k = 1, \ldots, n$, we denote by $\hat{f}_k$ the k-nearest neighbors estimator. We fix $k \in \{1, \ldots, n\}$; we construct, for every $i = 1, \ldots, n$ and $x \in \mathcal{X}$,

$$\hat{f}_k(\mathcal{D}_i)(x) := \sum_{j \neq i} W_j(x) Y_j \text{ where } W_j(x) = k^{-1} I(X_j \in J_k(x))$$

and $J_k(x)$ is the set of the k nearest neighbors of x among $\{X_j : j \neq i\}$. The estimator $\hat{f}_k(\mathcal{D}_i)$ is therefore the k-nearest neighbors estimator constructed from the observation set $\mathcal{D}_i$. One repeats the procedure for every $i = 1, \ldots, n$; one thus obtains n different estimators $\hat{f}_k(\mathcal{D}_1), \ldots, \hat{f}_k(\mathcal{D}_n)$. The VC risk of $\hat{f}_k$ is given by

$$r_{VC}(\hat{f}_k) = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \hat{f}_k(\mathcal{D}_i)(X_i))^2$$

One carries out this construction for every $k = 1, \ldots, n$; one thus obtains n VC risks: $\big(r_{VC}(\hat{f}_k)\big)_{k=1,\ldots,n}$. One then determines $\hat{k}$ such that $r_{VC}(\hat{f}_{\hat{k}}) = \min_{k=1,\ldots,n} r_{VC}(\hat{f}_k)$. The estimator obtained by cross-validation is therefore $\hat{f}_{\hat{k}}(\mathcal{D})$. It is a k-nearest neighbors estimator with a parameter $k = \hat{k}$ chosen empirically.

3.4 Exponential-weight methods

Unlike the cross-validation method, one resorts here to a partition of the data. One assumes that the estimators $\hat{f}_1, \ldots, \hat{f}_M$ have been constructed previously during the training phase. One now passes to the learning phase. In this phase, one may consider the learning sample as "frozen", and therefore the estimators $\hat{f}_1, \ldots, \hat{f}_M$ may be viewed as functions, now denoted $f_1, \ldots, f_M$. One therefore begins the learning phase with a set of M functions $\{f_1, \ldots, f_M\}$ and a set of n_V data points. For simplicity, we shall denote n_V by n.

For aggregation methods, the emphasis is placed on the learning phase. One begins this phase with a set of M functions $F := \{f_1, \ldots, f_M\}$ (which are our former estimators coming from the training phase, and which one assumes to be nonrandom for the second phase, the learning phase) and n observations, still denoted $(X_1, Y_1), \ldots, (X_n, Y_n)$, which one uses to construct different aggregation methods. The set F is called a *dictionary*.

The exponential-weight aggregation method is given by

$$\tilde{f} := \sum_{j=1}^{M} w_j f_j \text{ where } w_j := \frac{\exp(-nr(f_j)/T)}{\sum_{k=1}^{M} \exp(-nr(f_k)/T)} \tag{15}$$

where T is called the temperature parameter. Here one has $r(f) = n^{-1} \sum_{i=1}^{n} (Y_i - f(X_i))^2$.

This is the first example of an aggregation method that takes its values in the convex hull of F. One therefore does not choose a particular estimator in the dictionary, but rather tries to combine them in order to create an estimator as performant as the best one in F. We shall see this idea again many times in what follows.

3.5 Empirical risk minimization on the star-shaped class

We place ourselves here in the same framework as the previous section: one is given a dictionary $F = \{f_1, \ldots, f_M\}$ of M functions and n data points. One wants to construct aggregation procedures $\tilde{f}$ such that "with high probability,

$$R(\tilde{f}) \leq C \min_{f \in F} R(f) + r_n(M).$$

where $C \geq 1$ and $r_n(M)$ is the remainder."

The method of empirical risk minimization on the star-shaped class is carried out in two steps. One first constructs the empirical risk minimization method:

$$\tilde{f}^{MRE} \in \arg\min_{f \in F} r(f).$$

Then, one constructs the set $\hat{F}_1 := \mathrm{star}(F, \tilde{f}^{MRE}) = \{[\hat{f}^{MRE}, f] : f \in F\}$ where for every $f \in F$, $[\hat{f}^{MRE}, f] = \{\theta \hat{f}^{MRE} + (1-\theta) f\}$. The set $\hat{F}_1$ is a set constructed empirically. It is included in the convex hull of F and has the shape of a star with center $\tilde{f}^{MRE}$. Once $\hat{F}_1$ has been constructed, one defines the method of empirical risk minimization on the star-shaped class by

$$\tilde{f} \in \arg\min_{f \in \hat{F}_1} r(f).$$

3.6 Aggregation method by convexification of the dictionary

The most efficient aggregation methods take their values in the convex set of the dictionary $F = \{f_1, \ldots, f_M\}$. We shall denote this set by

$$\mathrm{conv}(F) := \Big\{ \sum_{i=1}^{M} \alpha_j f_j : \alpha_j \geq 0, \sum_{j=1}^{M} \alpha_j = 1 \Big\}.$$

An aggregation method that may then come naturally is the one that consists in minimizing the empirical risk on $\mathrm{conv}(F)$, that is,

$$\tilde{f}^C \in \arg\min_{f \in \mathrm{conv}(F)} r(f).$$

One may also preselect certain estimators that behave well empirically and construct an empirical risk minimizer on the convex hull of this set of estimators: one first constructs the empirical risk minimizer

$$\tilde{f}^{MRE} \in \arg\min_{f \in F} r(f).$$

One looks at the elements of F that have a small empirical risk:

$$\hat{F}_2 := \Big\{ f \in F : r(f) \leq \min_{f \in F} r(f) + \max\Big(\Big\| f - \tilde{f}^{MRE} \Big\|_n \sqrt{\frac{\log M}{n}}, \frac{\log M}{n} \Big) \Big\}$$

where $\|f\|_n^2 = n^{-1} \sum_{i=1}^{n} f(X_i)^2$ for every function f. One then constructs the convex hull associated with $\hat{F}_2$:

$$\mathrm{conv}(\hat{F}_2) := \Big\{ \sum_{f \in \hat{F}_2} \alpha_f f : \alpha_f \geq 0, \sum_{f \in \hat{F}_2} \alpha_f = 1 \Big\}.$$

Finally, the "preselection/convexification" method is given by

$$\tilde{f} \in \arg\min_{f \in \mathrm{conv}(\hat{F}_2)} r(f).$$

3.7 Bagging

The Bagging algorithm is a method combining two classical ideas in statistics: resampling (or *bootstrapping*) and the idea of regularization by averaging. This method was introduced in order to avoid a classical problem in learning: *overfitting*.

The overfitting problem can be understood in the following way. If F is a very rich class of functions, then the empirical risk minimizer on F:

$$\tilde{f}^{MRE} \in \arg\min_{f \in F} r(f)$$

may have a high risk. That is, the difference $R(\tilde{f}) - \inf_{f \in F} R(f)$ may be large. This may at first sight appear paradoxical, in the sense that $\mathbb{E}r(f) = R(f)$, so that by minimizing $r(f)$ over F one may expect the true risk R also to be close to the minimum over F. This intuition is true when F has small complexity, but becomes false when F has large complexity. In the latter case, it is not rare to find a function f in F such that $f(X_i) \simeq Y_i$ for every $i = 1, \ldots, n$, and therefore also $\tilde{f}^{MRE}(X_i) \simeq Y_i$ for every $i = 1, \ldots, n$. In this case, one will say that $\tilde{f}^{MRE}$ "fits the data too closely". This is clearly troublesome in a denoising model, in which such a situation amounts to not denoising the data but to returning approximately $Y_1, \ldots, Y_n$ (together with the noise) instead of seeking to estimate $f(X_1), \ldots, f(X_n)$.

To avoid the overfitting problem of certain estimators (such as the empirical risk minimizer on complex models), the Bagging method was introduced. Suppose that we have an estimator $\hat{f}$. One is given two integers $n' \leq n$ and m. The first step is to construct "resampled" samples, that is, sets $\mathcal{D}_1, \ldots, \mathcal{D}_m$ of data, each of them constructed by choosing uniformly at random and with replacement from $\{(X_1, Y_i), \ldots, (X_n, Y_n)\}$. Once these samples have been constructed, one produces the different estimators $\hat{f}(\mathcal{D}_1), \ldots, \hat{f}(\mathcal{D}_m)$, which represent the same estimator $\hat{f}$ constructed from the different data sets $\mathcal{D}_1, \ldots, \mathcal{D}_m$. The bagging estimator is therefore

$$\tilde{f}^{bagging} := \frac{1}{m} \sum_{j=1}^{m} \hat{f}(\mathcal{D}_j).$$

In binary classification, one will take

$$\tilde{f}^{bagging} := \begin{cases} 1 & \text{if } \frac{1}{m} \sum_{j=1}^{m} \hat{f}(\mathcal{D}_j) \geq 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

3.8 Adaboost

The Adaboost method ("Adaptive Boosting") is the Boosting algorithm most used in practice. A Boosting algorithm is a recursive aggregation method on a large dictionary. It is defined in the binary classification model.

Consider the binary classification model and denote by $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ our learning set on a space $\mathcal{X}$. For simplicity, we shall assume that the outputs $Y_1, \ldots, Y_n$ take values in $\{-1, 1\}$. One is given a class F of functions from $\mathcal{X}$ into $\{-1, 1\}$. One seeks to construct a procedure that performs as well as the best element in $\mathrm{vect}(F)$, the vector space spanned by F. This problem is called the *linear aggregation problem*. The elements of F are sometimes called *weak estimators*.

To present the AdaBoost algorithm, we introduce a few notations. Every vector $w \in \mathbb{R}^n$ such that $w_i \geq 0$ for $i = 1, \ldots, n$ and $\sum_{i=1}^{n} w_i = 1$ may be viewed as a probability measure on the n-element set $\mathcal{D} = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$. The AdaBoost algorithm consists in modifying the weight vector w by assigning more and more mass to the data that are "difficult to classify by the current weak estimator". One is given a number m of iterations. AdaBoost is constructed in the following way:

1. One initializes the weights uniformly:

$$w_i = 1/n, \quad \forall i = 1, \ldots, n.$$

2. for $j = 1, \ldots, m$:

   - Find $f_j \in \arg\min_{f \in F} \sum_{i=1}^{n} w_i I(f(X_i) \neq Y_i)$ and write $e_j := \sum_{i=1}^{n} w_i I(f_j(X_i) \neq Y_i)$.

- Compute $\alpha_j := \frac{1}{2}\log\big(\frac{1-e_j}{e_j}\big)$.
- Update of the weights: for every $i = 1, \ldots, n$ such that $f_j(X_i) \neq Y_i$, $w_i \leftarrow w_i \frac{1-e_j}{e_j}$.
- Normalize the weights: $c := \sum_{i=1}^n w_i$, $w_i \leftarrow w_i/c$.

3. The AdaBoost algorithm is

$$\tilde{f} := \mathrm{sign}\Big(\sum_{j=1}^m \alpha_j f_j\Big).$$

The AdaBoost algorithm can be viewed as a gradient descent algorithm. Gradient descent methods are used to find the minimum of functions of several variables. Let $f : \mathbb{R}^d \to \mathbb{R}$. One assumes f differentiable and one denotes by $\nabla f(x)$ the gradient of f at x. One fixes $\epsilon > 0$. A gradient descent method is written recursively as:

1. One is given an initial point $x_0 \in \mathbb{R}^d$.
2. At step k, one performs the stopping test $\|\nabla f(x_k)\|_2 \leq \epsilon$.
3. Determine $\alpha_k \in \arg\min_{\alpha>0} f(x_k - \alpha \nabla f(x_k))$.
4. New iterate: $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$.

The idea behind this method is that, locally at $x \in \mathbb{R}^d$, the direction of steepest descent for f is $-\nabla f(x)/\|\nabla f(x)\|_2$. Indeed, one can see that for every $v \in \mathbb{R}^d$ such that $\|v\|_2 = 1$, one has

$$f(x+v) = f(x) + \big\langle \nabla f(x), v\big\rangle + o(1)$$

and then the minimum $\min_{\|v\|_2=1} f(x+v)$ is approximately attained by the minimizing argument of $\min_{\|v\|_2=1}\big\langle \nabla f(x), v\big\rangle$, that is, $-\nabla f(x)/\|\nabla f(x)\|_2$. At each step of a gradient descent method, one is therefore led to compute

$$(\alpha_k, d_k) \in \arg \min_{(\alpha,d)\in\mathbb{R}^+\times\mathcal{S}_2^{d-1}} f(x_k - \alpha d) \tag{16}$$

where $\mathcal{S}_2^{d-1}$ is the unit sphere of $(\mathbb{R}^d, \ell_2^d)$. One then obtains the new iterate: $x_{k+1} = x_x - \alpha_k d_k = x_k - \alpha_k \nabla f(x_k)$. Finally, the parameter α_k is called the *step size* of the method; there exist many other ways of choosing the step size.

It is possible to view AdaBoost as a gradient descent method in which the descent directions are taken in F and the step size is chosen as in step 3 of the previous gradient descent algorithm. Indeed, the AdaBoost classifier is the sign of a linear combination of elements of F. Before taking the sign of this linear combination, one wishes to perform a gradient descent for a certain function inside $\mathrm{vect}(F)$. The AdaBoost method is in fact an m-step gradient descent method used for the minimization problem

$$\min_{f\in\mathrm{vect}(F)} r(f) \text{ where } r(f) := \frac{1}{n}\sum_{i=1}^n \exp(-Y_i f(X_i)). \tag{17}$$

One recognizes here the empirical risk in classification $r(\cdot)$ associated with the loss function $\ell((x,y),f) = \phi(-y(f(x))$ for $\phi(u) = \exp(u)$. The function ϕ is a classical example of a convex relaxation of the Heaviside loss $\phi_0(u) = \mathbb{1}_{u\geq 0}$ in classification.

Theorem 3.1 *The AdaBoost algorithm is an m-step gradient descent method on the function $r(\cdot)$ in which, at each iteration j, a weak estimator $f_j \in F$ is chosen in the direction (restricted to F) of steepest descent of $r(\cdot)$ at the previously constructed point h_{j-1}. The step size α_j is chosen so as to minimize the function $r(\cdot)$ along the line $\alpha \to h_{j-1} + \alpha f_j$. In other words, AdaBoost is the following iterative algorithm:*

1. *At the first step, one defines*

$$(\alpha_1, f_1) \in \arg \min_{(\alpha,f)\in\mathbb{R}\times F} r(\alpha f).$$

2. *At step $j = 2, \ldots, m$, one has a function of the form $h_{j-1} = \sum_{k=1}^{j-1} \alpha_k f_k$. One then seeks*

$$(\alpha_j, f_j) \in \arg \min_{(\alpha,f)\in\mathbb{R}\times F} r(h_{j-1} + \alpha f).$$

3. *Finally, Adaboost is the classifier* $\operatorname{sign}\Big(\sum_{k=1}^{m} \alpha_k f_k \Big)$.

Proof: We place ourselves at step j. One therefore assumes h_{j-1} constructed and one studies $r(\cdot)$ locally at h_{j-1}. The direction of steepest descent among F of $r(\cdot)$ at h_{j-1} is given by the function $f \in F$ minimizing the derivative of $w \to r(h_{j-1} + wf)$ at $w = 0$. In other words, one has

$$f_{j+1} \in \arg \min_{f\in F} \partial_w r(h_{j-1} + wf)(0).$$

Differentiating, one obtains

$$\begin{aligned}
\arg \min_{f\in F} \partial_w r(h_{j-1} + wf)(0) &= \arg \min_{f\in F} \sum_{i=1}^{n} -Y_i f(X_i) \exp\big(- Y_i h_{j-1}(X_i)\big) \\
&= \arg \min_{f\in F} \sum_{i=1}^{n} (2I(f(X_i) \neq Y_i) - 1) \exp\big(- Y_i h_{j-1}(X_i)\big) \\
&= \arg \min_{f\in F} \sum_{i=1}^{n} I(f(X_i) \neq Y_i) \exp\big(- Y_i h_{j-1}(X_i)\big)
\end{aligned}$$

One chooses the step size α_j minimizing $r(\cdot)$ along the line $\alpha \to h_{j-1} + \alpha f_j$. One has

$$\begin{aligned}
\partial_\alpha r(h_{j-1} + \alpha f_j) &= \frac{1}{n} \sum_{i=1}^{n} (-Y_i f_j(X_i)) \exp(-\alpha Y_i f_j(X_i)) \exp(-Y_i h_{j-1}(X_i)) \\
&= \frac{1}{n} \sum_{i=1}^{n} \Big(- \exp(-\alpha) I(f_j(X_i) = Y_i) + \exp(\alpha) I(f_j(X_i) \neq Y_i)\Big) \exp\big(- Y_i h_{j-1}(X_i)\big).
\end{aligned}$$

Hence the minimum is attained at

$$\alpha_j = \frac{1}{2} \log\Big(\frac{\sum_{i=1}^{n} I(f_j(X_i) = Y_i) \exp\big(- Y_i h_{j-1}(X_i)\big)}{\sum_{i=1}^{n} I(f_j(X_i) = Y_i) \exp\big(- Y_i h_{j-1}(X_i)\big)} \Big).$$

□

References

[AC09] J.-Y. Audibert and O. Catoni. Risk bounds in linear regression through pac-bayesian truncation. Preprint, arXiv:0902.1733, 2009.

[Aka73] H. Akaike. Information theory and an extension of the maximum likelihood principle. In B. N. Petrov and F. Csaki, editors, *2nd International Symposium on Information Theory*, pages 267–281. Budapest: Akademia Kiado, 1973.

[BRT10] P. J. Bickel, Y. Ritov, and A. Tsybakov. Simultaneous analysis of lasso and dantzig selector. To appear in the Annals of Statistics, 2010.

[BTW07] F. Bunea, A.B. Tsybakov, and M.H. Wegkamp. Aggregation for Gaussian regression. *Annals of Statistics*, 35:1674–1697, 2007.

[EHJT04] B. Efron, T. Hastie, I. Johnstone, and R. Tibshirani. Least angle regression. *The Annals of Statistics*, 32(2):407–499, 2004.

[GKKW04] L. Györfi, M. Kohler, A. Krzyżack, and H. Walk. *A distribution-free theory of nonparametric regression*. Springer, 2004.

[HK70] A. E. Hoerl and R. W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12, 1970.

[HTF09] T. Hastie, R. Tibshirani, and J. Friedman. *The elements of statistical learning*. Springer, 2nd edition, 2009.

[HUL93] Jean-Baptiste Hiriart-Urruty and Claude Lemaréchal. *Convex analysis and minimization algorithms. II*, volume 306 of *Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. Springer-Verlag, Berlin, 1993. Advanced theory and bundle methods.

[Mal73] C. L. Mallows. Some comments on c_p. *Technometrics*, 15:661–676, 1973.

[Sch78] G. Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6:461–464, 1978.

[Tib96] R. Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society B*, 58(1):267–288, 1996.